DATAMART CONSOLIDATION & KPI HARMONIZATION

Scattered datamarts and DWHs consolidated into a conformed layer, with one KPI dictionary and usage-based decommissioning

Start with a consolidation assessment, consolidate in waves on an existing platform, then run the conformed layer every month

1 **The challenge**
Symptoms, from assessment to consolidation, what the package does. p. 2 – 4

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Symptoms of Scattered Datamarts

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Datamarts are usually built one by one, for one unit or one need at a time. After some years the same data, the same KPIs and the same jobs exist in several places.

01 · THE CHALLENGE

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Many datamarts built over years

- Each unit built its own datamart
- The same data loaded several times
- No complete list of what exists

Inventory

One KPI, several answers

- The same KPI with different formulas
- Numbers differ between reports
- Time spent explaining differences

KPI conflicts

Objects no one uses

- Tables and columns kept just in case
- Reports with no recent users
- Jobs that still run every night

Unused objects

Rising maintenance cost

- Hundreds of ETL jobs to maintain
- Licences on several platforms
- Each change tested in many places

Cost

No common model

- Dimensions defined differently
- No shared business keys
- Different history rules per datamart

Model

Slow change, long runs

- A new need touches several datamarts
- Long batch windows
- Knowledge held by a few people

Change

WHAT CONSOLIDATION CHANGES

One conformed layer
shared dimensions and facts

One KPI dictionary
one agreed definition per KPI

Fewer objects
unused ones decommissioned

Reconciled numbers
old and new compared per wave

Common standards
for modelling and ETL

From Assessment to Consolidation

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The package is the next step after a DWH Assessment. Typical findings of the assessment map to consolidation steps; the scope boundary keeps the work on the current platform.

02 · FROM ASSESSMENT TO CONSOLIDATION

TYPICAL DWH ASSESSMENT FINDING	CONSOLIDATION STEP IN THIS PACKAGE
Scattered datamarts built over the years	Inventory and conformed layer; datamarts rebuilt on shared dimensions and facts
The same KPI calculated in several ways	KPI dictionary; one definition per KPI, agreed with its owner
Tables, columns and reports no one uses	Query-log usage analysis; decommission with owner approval
Different numbers for the same question	Old and new numbers reconciled in a parallel run per wave
High maintenance and licence cost	Fewer objects, jobs and platforms to run after each wave

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SCOPE BOUNDARIES

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Data Modernization
This package consolidates onto a platform the customer already runs. A move to a new platform, for example to cloud, is Data Modernization.

DWH Assessment
Evaluates the whole platform. This package carries out the consolidation found there; its own assessment (tier A) covers consolidation only.

Data governance
The KPI dictionary here covers reporting KPIs. An enterprise glossary and an ownership framework belong to data governance.

Regulatory reporting
Regulatory datamarts can be in scope as objects. Regulatory report families belong to the Regulatory Reporting Data Layer package.

What the Package Does

KOMTAS

Datamart Consolidation & KPI Harmonization brings scattered datamarts and DWHs onto one conformed layer, agrees one definition per KPI and removes what is no longer used.

03 · WHAT IT DOES

THREE RESULTS

Conformed layer
Conformed dimensions, shared facts, common business keys and history rules; subject-area marts rebuilt on it.

KPI dictionary
Definition, formula, owner, source and grain per KPI; conflicts between datamarts resolved.

Decommission
Unused, duplicated and low-use tables, columns, reports and datamarts removed, based on query-log usage and owner approval.

FOR WHOM

Organisations with several datamarts or DWHs built over the years, for example in banking, telecom, retail and insurance. «DIGITALLENCY DOĞRULASIN»

Consolidation on an existing platform
The target is one of the platforms the customer already runs. A move to a new platform, for example to cloud, is handled in the Data Modernization service.

WHEN IT IS USEFUL «DIGITALLENCY DOĞRULASIN»

- Findings of a DWH Assessment
- Conflicting numbers between reports
- Rising maintenance or licence cost
- Many datamarts built over many years
- A merger of units or companies
- End of life of one of several platforms

OUT OF SCOPE «DIGITALLENCY DOĞRULASIN»

- Migration to a new platform (Data Modernization)
- Changes to source systems
- New report design beyond repointing
- Data cleansing in source systems
- Licence procurement

Approach and Architecture

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A generic, platform-independent approach. Existing datamarts are rebuilt on a conformed layer, reports are repointed, and the objects they replace are archived and removed.

04 · ARCHITECTURE

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Sources

operational systems

- Core systems
- Files and feeds
- External data

Existing DWHs and datamarts

built over the years

- Several platforms
- Overlapping datamarts
- Own ETL per datamart

Conformed layer

one model on an existing platform

- Conformed dimensions
- Shared facts
- Common business keys
- History rules

Subject-area marts

rebuilt on the conformed layer

- Per business area
- Agreed KPIs
- Consumer views

Reports and universes

repointed to the new marts

- BI reports
- BI universes
- Extracts and feeds

Decommissioned objects

archived, then removed

- Unused tables
- Replaced datamarts
- Old ETL jobs

CROSS-CUTTING

KPI dictionary

One definition per KPI, used by every mart.

Modelling and ETL standards

Naming, conformed dimension rules, ETL patterns, reviews.

Query-log usage analysis

Usage of tables, columns and reports from DWH logs.

Reconciliation

Old and new numbers compared per KPI.

Parallel run per wave

Old and new run side by side until accepted.

Built on an existing platform. The conformed layer is built on a platform the customer already runs «PLATFORM». A move to a new platform is Data Modernization, a separate service.

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KPI Harmonisation

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Each KPI gets one agreed definition, one owner and one implementation on the conformed layer. Variants stay only where the business needs them, and then under their own name.

05 · KPI HARMONISATION

HARMONISATION STEPS

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1 Collect KPIs
From reports, universes and datamart code.

2 Compare variants
How each datamart calculates the KPI.

3 Decide with the owner
The business owner agrees one definition.

4 Build once
Formula built on the conformed layer, at the agreed grain.

5 Reconcile
New results compared with the current reports.

KPI DICTIONARY ENTRY · ILLUSTRATIVE EXAMPLE

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KPI	Active customers
Definition	Customers with at least one qualifying activity in the period
Formula · grain	Count of distinct customers · month, branch, segment
Owner · source	Business unit owner · conformed layer: customer and transaction facts
Calculation per datamart	Datamart 1: a transaction in the last 3 months · Datamart 2: a product with a balance · Datamart 3: a login in the last 30 days
Decision	One definition agreed by the owner; the Datamart 3 logic is kept as a named variant, "digital active customers".

Example for illustration only. Template columns: definition, formula, owner, source, grain, variants, calculation per datamart, decision.

KPI definitions are decided by the business. Komtaş prepares the conflict table and the options; the KPI owner decides. Each decision is recorded in the KPI dictionary.

Decommission Based on Usage

KOMTAS

What can be removed is decided from query logs, not from opinions. Each candidate is confirmed by its owner, archived, and only then removed.

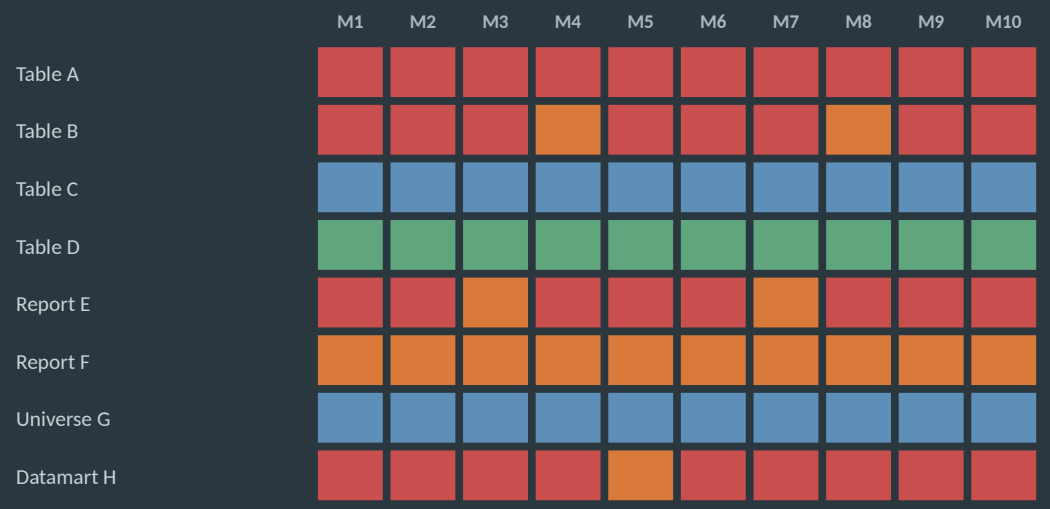
06 · DECOMMISSION

DECOMMISSION STEPS

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1. **Collect query logs** for an agreed time window
2. **Measure usage** per table, column and report
3. **Classify** unused, duplicated, low use
4. **Confirm with owners** each candidate approved or kept
5. **Archive** data and definitions kept retrievable
6. **Remove** objects, jobs and reports switched off

USAGE HEAT MAP · ILLUSTRATIVE



Unused
No query in the log window: candidate to remove.

Low use
Few users: candidate to merge or move.

Duplicated
Same content elsewhere: one copy kept.

The log window matters. Objects used only at year end or in audits can look unused in a short window. The window is agreed with the owners, and every candidate is confirmed before removal. «DIGITALLENCY DOĞRULASIN»

Accelerators

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Four accelerators taken from earlier consolidation and migration work. They shorten the analysis, keep waves consistent and reduce hand-written code.

07 · ACCELERATORS

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1

Query-log decommission analysis

Usage of tables, columns and reports from DWH query logs.

- Usage frequency per object
- Unused, duplicated, low-use lists
- Owner confirmation per item
- Suited to large log volumes

Used at a private bank, in a big data environment

2

KPI dictionary template

One entry per KPI, with a column for each datamart's calculation.

- Definition, formula, grain
- Owner and source
- Calculation per datamart
- Decision and variants

Used in a private bank consolidation

3

Modelling and ETL standards

Rules applied to every wave, with a committee rhythm.

- Naming conventions
- Conformed dimension rules
- ETL patterns
- Code review checklist

Modelling and ETL standards committees

4

Rule-based migration framework

Moves objects by definitions, without hand-written code.

- Object definitions as input
- Migration code generated
- Option for high object counts
- Repeatable per wave

A migration framework used at a private bank «DIGITALLENCY DOĞRULASIN»

Standards as governance. A modelling committee and an ETL committee meet on a regular rhythm; the model and code of each wave are reviewed against the standards before release.

Three Service Tiers: Assessment, Consolidation, Run

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The assessment maps datamarts, usage and KPI conflicts. Consolidation builds the foundation once and then works in waves. Run keeps the conformed layer and the KPI dictionary current.

08 · SERVICE TIERS

A — Consolidation Assessment

fixed fee «...» · 4 – 8 weeks

- Inventory: datamarts, tables, jobs, reports
- Query-log usage analysis
- Decommission candidate list
- KPI conflict analysis
- Conformed model draft, wave plan

OUTPUT

Inventory · usage heat map · decommission candidates · KPI conflict table · conformed model draft · wave plan and B estimate

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B — Consolidation

foundation once · fee «...» per wave

- Standards, conformed core, KPI dictionary
- Reconciliation framework
- Per wave: model, ETL, repointing
- Parallel run and decommission
- Handover per wave

OUTPUT

Datamarts rebuilt on the conformed layer · KPIs reconciled · reports repointed · replaced objects removed

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C — Run

monthly fee «...» · minimum term 12 months

- New needs on the conformed layer
- KPI dictionary maintenance
- Standards compliance checks
- Remaining decommission steps
- Unused hours are not carried over

OUTPUT

Monthly report · KPI dictionary kept current · new requirements from a monthly capacity

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Assessment first. Consolidation starts from the assessment, or from an equivalent analysis already done by the customer «A REF». «DIGITALLENCY DOĞRULASIN»

Assessment: Work Flow and Timeline

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Five phases over four to eight weeks. The assessment ends with a decommission candidate list, a conformed model draft, a wave plan and an estimate for consolidation.

09 · ASSESSMENT

INDICATIVE PLAN · BAND M · 6 WEEKS

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SIZE DRIVERS

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Datamarts | Tables | ETL jobs | Reports | KPIs | Source DWHs

Each driver falls into band S, M or L. The size is the highest band reached by at least three drivers. Reports include BI universes.

DELIVERABLES

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- Inventory: datamarts, tables, ETL jobs, reports, universes, KPIs, owners
- Usage heat map per table, column and report
- Decommission candidate list
- KPI conflict table
- Conformed model draft
- Wave plan and consolidation estimate

Duration by size. Small about 4, medium about 6, large about 8 weeks. Estates beyond L are planned as a programme. «DIGITALLENCY DOĞRULASIN»

Wave Delivery and Parallel Run

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The foundation is built once. Datamarts then move to the conformed layer in waves; each wave runs in parallel with the current datamarts until the numbers agree.

10 · CONSOLIDATION DELIVERY

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1 — Standards

Naming, conformed dimension rules, ETL patterns.

2 — Conformed core

Shared dimensions, KPI dictionary, reconciliation set-up.

3 — Model and ETL

Wave datamarts rebuilt on the conformed layer.

4 — Reconciliation

Old and new KPI values compared.

5 — Repointing, parallel run

Reports and universes repointed; both run.

6 — Decommission, handover

Replaced objects removed; runbook handed over.

FOUNDATION · ONCE

PER WAVE

PARALLEL RUN

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ACCEPTANCE PER WAVE

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- Agreed KPIs within tolerance «NUMBER»
- Tolerance met for «NUMBER» periods
- Repointed reports accepted by owners
- Decommissioned objects approved by owners

Wave size. A wave covers 2–4 datamarts or one business area. Current datamarts keep running until the wave is accepted; replaced objects are then removed. «DIGITALLENCY DOĞRULASIN»

What the Customer Provides

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Knowledge of the KPIs and of the current datamarts sits with the customer. The work needs time from four groups, and three conditions decide when it can start.

11 · CUSTOMER INPUTS

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Business KPI owners

per business area

- Meaning of each KPI
- Decision on conflicts
- Approval of decommission
- Review of reconciliation

DWH team

current datamarts

- Datamart and ETL knowledge
- Access to query logs
- Model and code reviews
- Takeover after each wave

BI team

reports and universes

- Report and universe list
- Repointing tests
- Report owner contacts
- Acceptance of repointed reports

IT operations

platform and access

- Environments on the platform
- Scheduling and operations
- Archive and removal steps
- Security approvals

CONDITIONS TO START

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Access to query logs
Logs kept for a period long enough to show real usage.

Named KPI owners
An owner who can decide for each KPI in scope.

An existing target platform
If the target is a new platform, the work is Data Modernization.

Decisions stay with the business. Komtaş prepares the analysis and the options. KPI definitions and decommission approvals are decided by the owners on the customer side.

Experience

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Reference work on datamart and DWH consolidation, shown by sector.

REFERENCE WORK

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Sector	Work	Scope	Year
A private bank	**Consolidation of 10+ datamarts**	Datamarts developed since 2007 consolidated. Modelling and ETL standards committees; query logs analysed in a big data environment to find tables and columns to decommission; KPI dictionary; 4 parallel agile teams; deployment of 30 datamarts coordinated.	«YEAR»
A telecom operator	**3 DWHs into one**	MSSQL, Oracle and Sybase DWHs consolidated into one DWH: 240 TB, 12,000 tables, ~540 SSIS jobs, ~1,000 T-SQL scripts, ~200 BO universes, ~4,000 tables replicated with Golden Gate.	«YEAR»

MEASURED RESULTS · TELECOM OPERATOR

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CDR processing

20 hours → **1 hour**

Campaign customer group selection

24 hours → **40 minutes**

WHAT THE REFERENCES SHOW

Standards as governance
Modelling and ETL committees kept many datamarts on one set of rules.

Usage before removal
Query logs, analysed at scale, showed which tables and columns to remove.

Large estates in one DWH
Three DWHs with 12,000 tables consolidated into one DWH.

References are shown by sector, without customer names.

About Komtaş, and the Next Step

Who delivers Datamart Consolidation & KPI Harmonization, and what is needed to start.

KOMTAS

12 · ABOUT KOMTAŞ AND NEXT STEP

WHY KOMTAŞ

Google Cloud Premier partner
Komtaş is a Google Cloud Premier partner.

Microsoft Azure partner
Komtaş is also a Microsoft Azure partner.

Data warehouse and integration work
Data warehouse, data integration and BI projects in several sectors. «DIGITALLENCY DOĞRULASIN»

Consolidation experience
Reference work by sector on p. 13. «DIGITALLENCY DOĞRULASIN»

From assessment to monthly run
Assessment, consolidation and Run from one partner.

Since 1989, focused on data · 250+ specialists · offices in Istanbul, Ankara, Izmir and Dubai

NEXT STEPS

1. **Walkthrough** — today — you are here
2. **Discovery call** — datamarts, KPIs, query logs, platform
3. **Assessment proposal** — fixed fee, scope and plan
4. **Kick-off** — access, inventory starts

Useful for the discovery call. A list of DWHs and datamarts with their platforms; the ETL and BI tools in use; known KPI conflicts between reports; whether query logs are kept, and for how long; DWH Assessment results, if available.